Understanding Myeloma Basics

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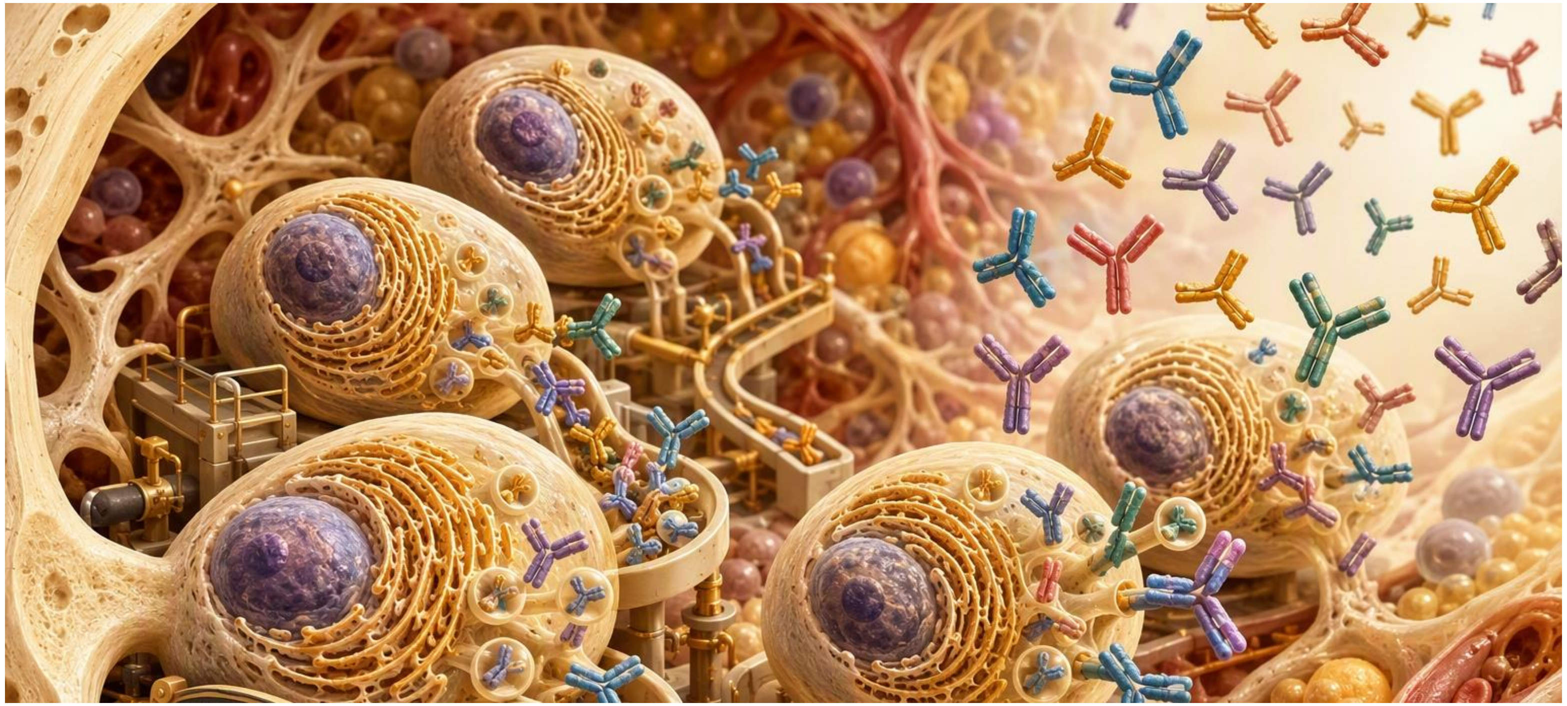
How Healthy Plasma Cells Work

Your body's antibody factories

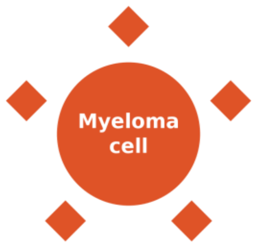


How Myeloma Starts

One cell goes wrong — and copies itself

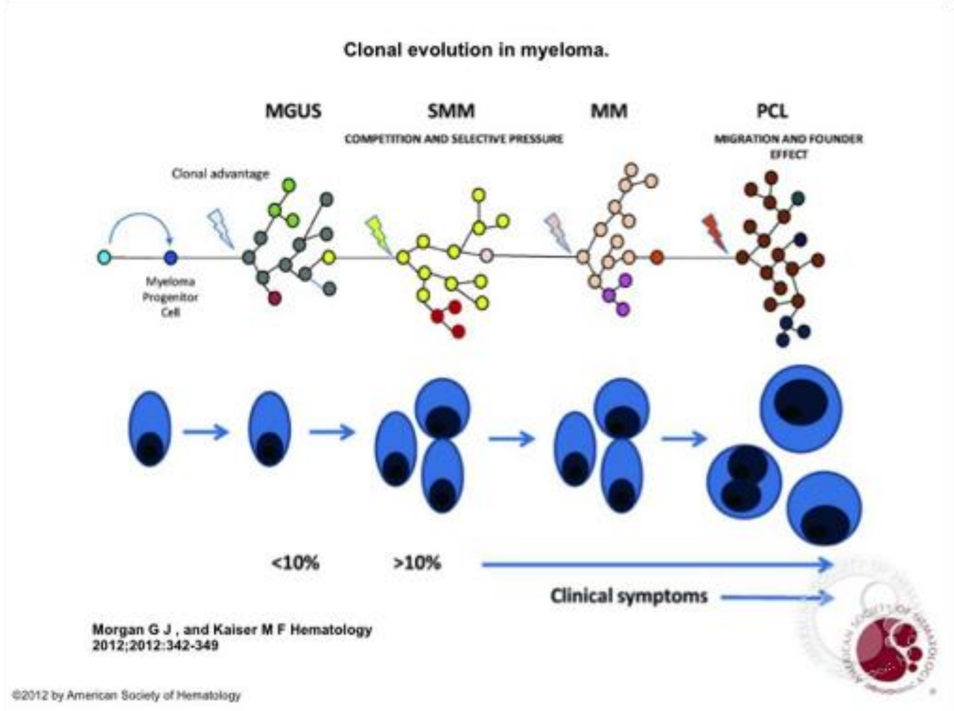
Many different antibody shapes

one plasma cell per germ — thousands of different defenders



One repeated abnormal protein

copied over and over — a single mistake, endlessly reprinted



One clone → one abnormal protein, an M-protein

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MGUS → Smoldering → Active Myeloma

MGUS

Common finding

M-protein <3g/dL and
Bone marrow plasma cells <10%

No Myeloma-defining Events

SMOLDERING

Common finding

M-protein ≥3g/dL or
Bone marrow plasma cells 10-59%

No Myeloma-defining Events

ACTIVE

Bone marrow plasma cells ≥10% and

Myeloma-defining Event

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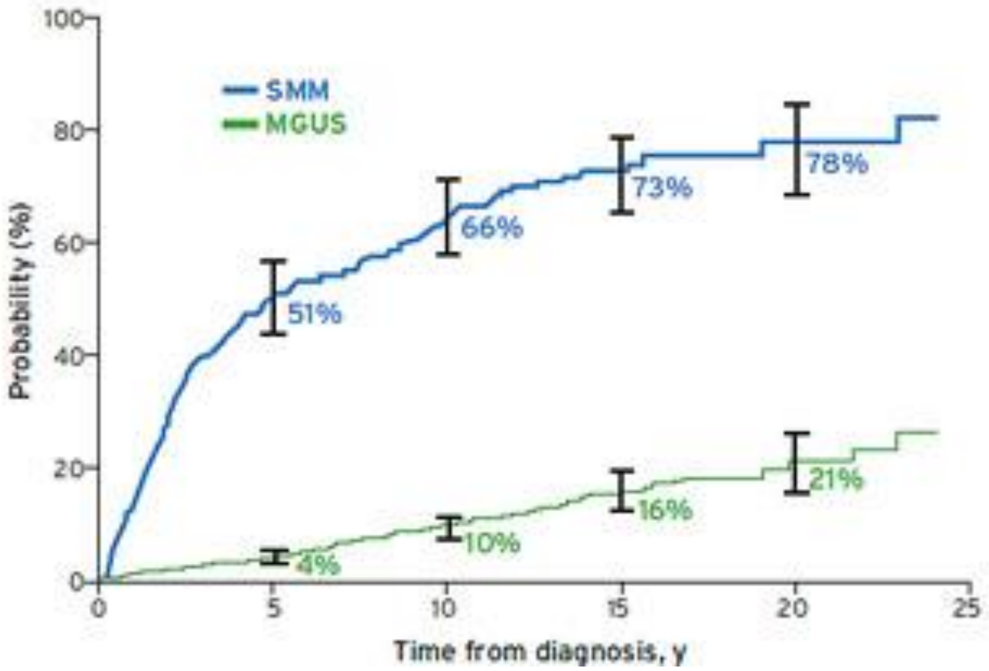
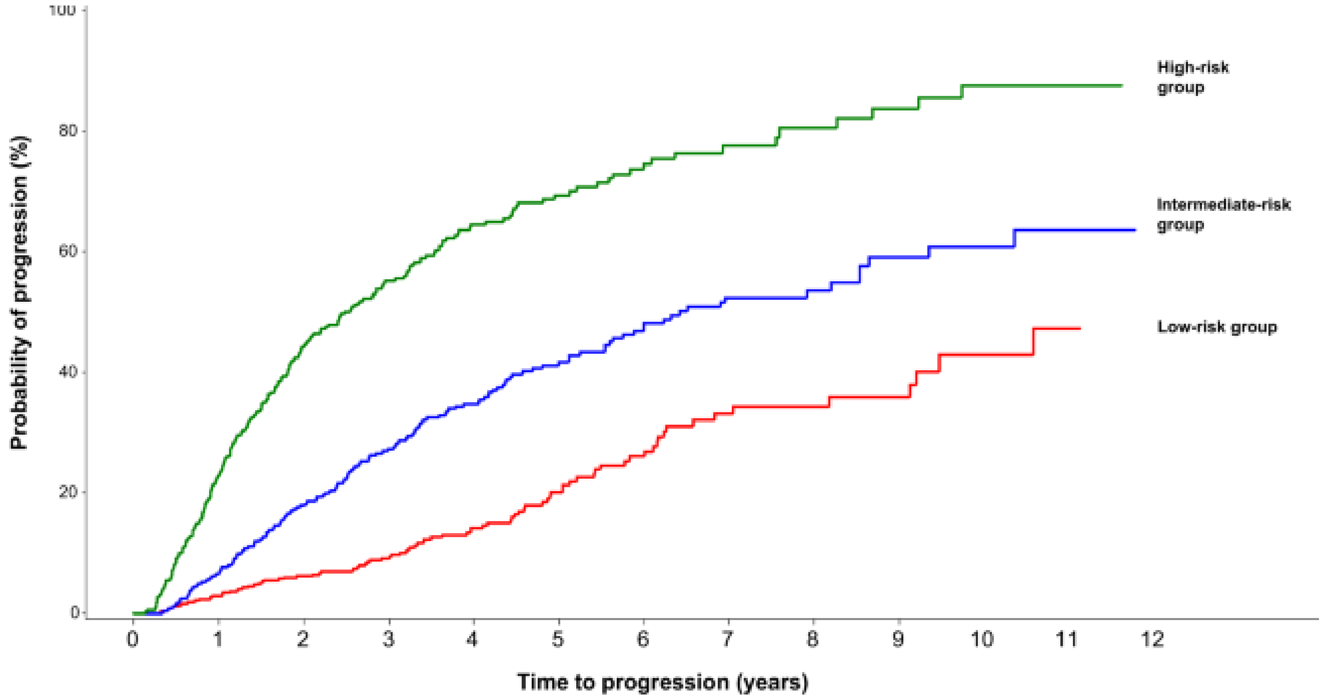
Active Surveillance, Explained

Why your doctor may watch, not treat





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CRAB and SLiM = Myeloma-Defining Events

C - Calcium

Elevated Blood Calcium: Broken down bone tissue releases calcium into the bloodstream, causing thirst, nausea, constipation, confusion.

R - Renal

Kidney Damage: Excess light chains can damage the kidney, revealed by rising blood creatinine levels. Kidney failure can cause nausea, tiredness, swelling.

A - Anemia

Low Red Blood Cells: Crowded marrow makes fewer red cells, leading to unexpected fatigue, weakness, or shortness of breath.

B - Bone

Lytic Lesions: Weakening bone areas (Swiss-cheese holes) detected on X-rays, CT, or PET/CT scans that lead to bone fractures and pain.

S - Sixty

Sixty % or more Plasma Cells: A high number of plasma cells on bone marrow biopsy.

Li – Light Chain Ratio

Involved Free Light Chain Ratio ≥100: A very high free light chain ratio.

M - MRI

More than 1 Bone Marrow Lesion: Whole body MRIs can detect subtle changes in the bone marrow, even when CT or PET/CT scans were normal.

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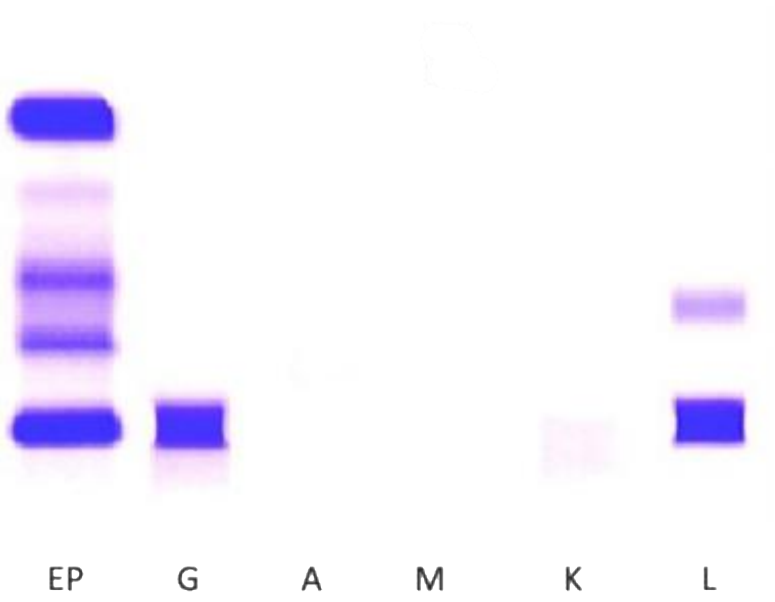
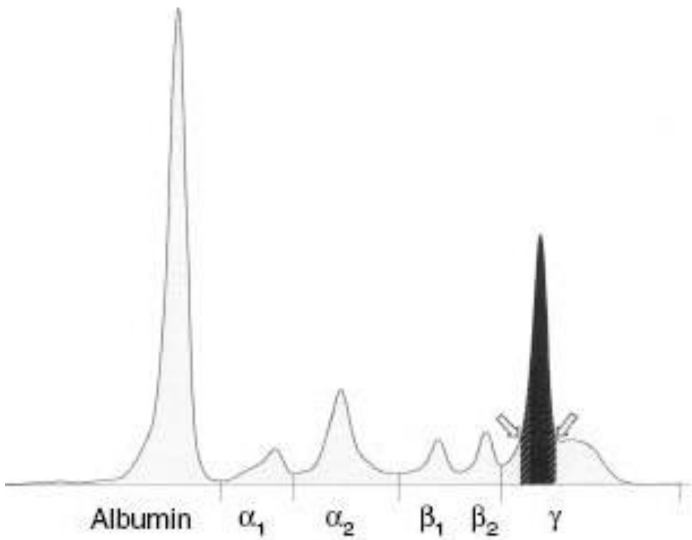
Measuring the Abnormal Protein

Serum or Urine Protein Electrophoresis and Immunofixation (M-Protein)



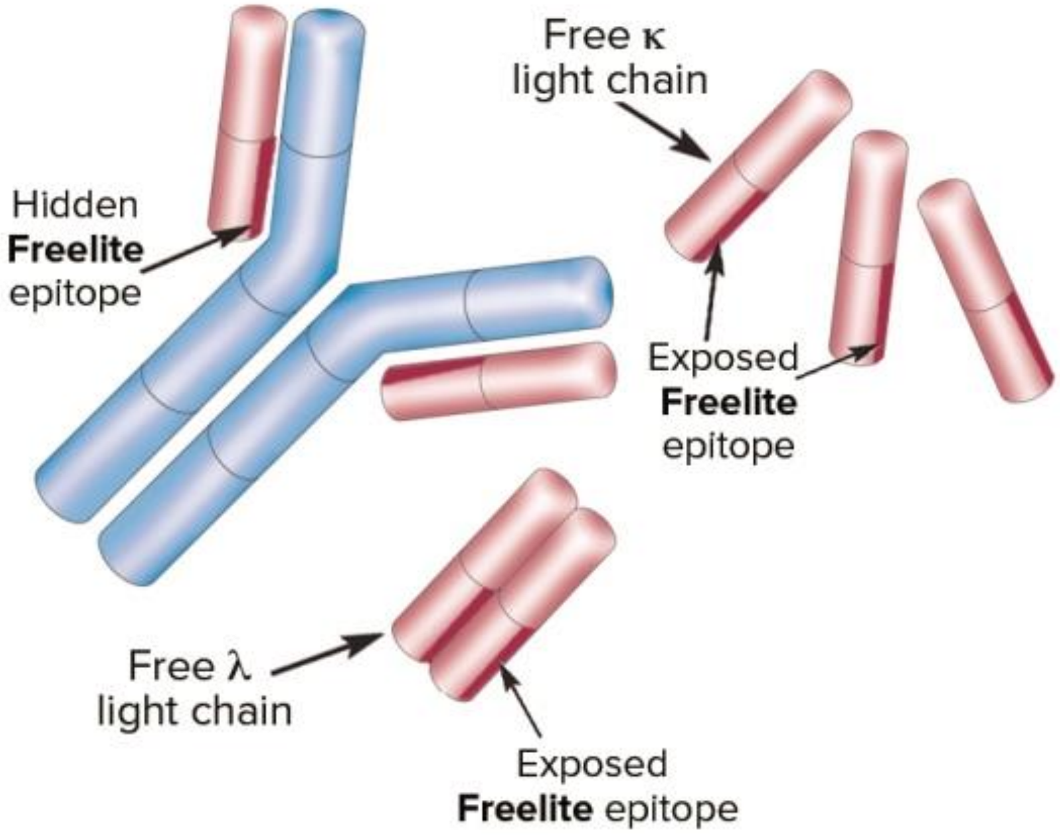


Total protein: 7.8 g/dL (6.0–8.3)
A/G: 0.81

Fractions	%	g/dL		Ref. g/dL
Albumin	44.9	3.50		3.43–4.84
α_1	4.7	0.37		0.21–0.44
α_2	13.0	1.01	H	0.54–0.97
$\beta_{1\ \&\ 2}$	9.5	0.74		0.65–1.03
γ	27.9	2.18	H	0.70–1.47
M-Spike	18.8	1.47		

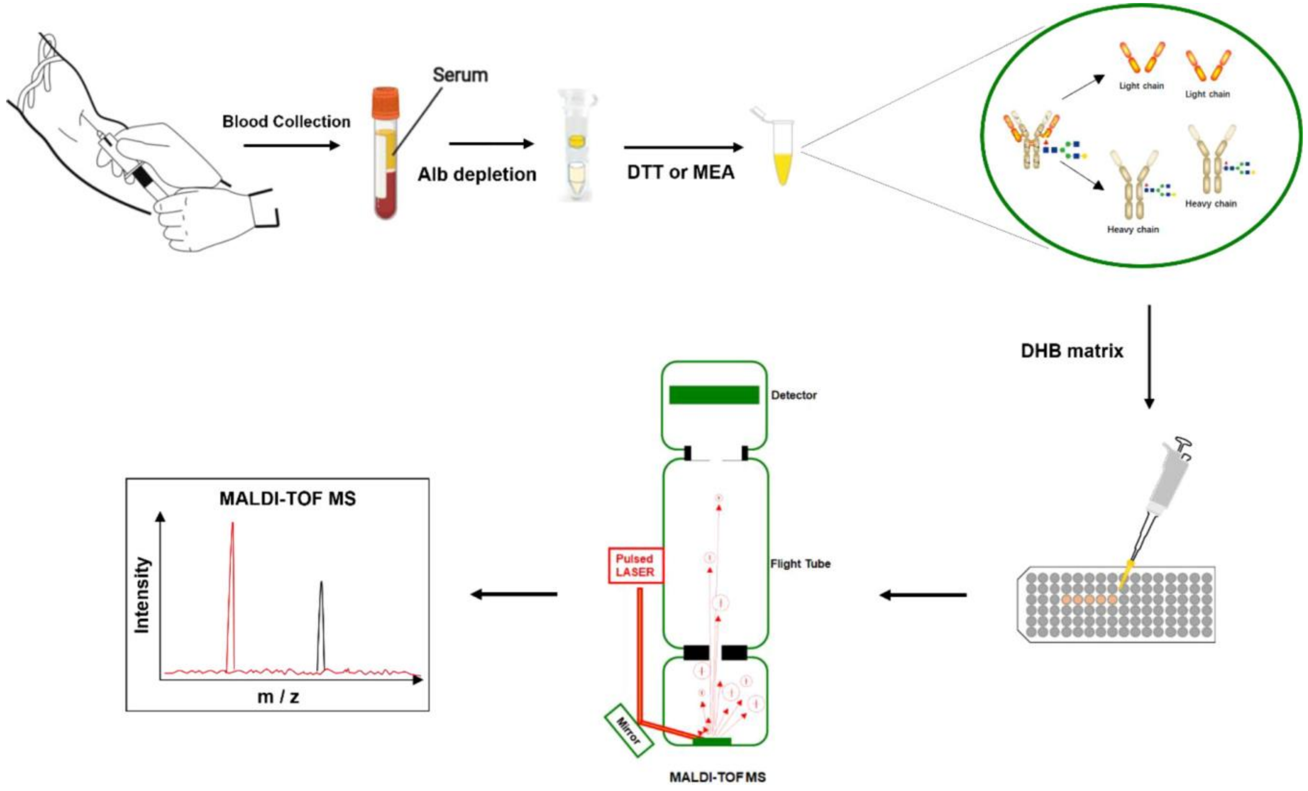
Measuring the Abnormal Protein

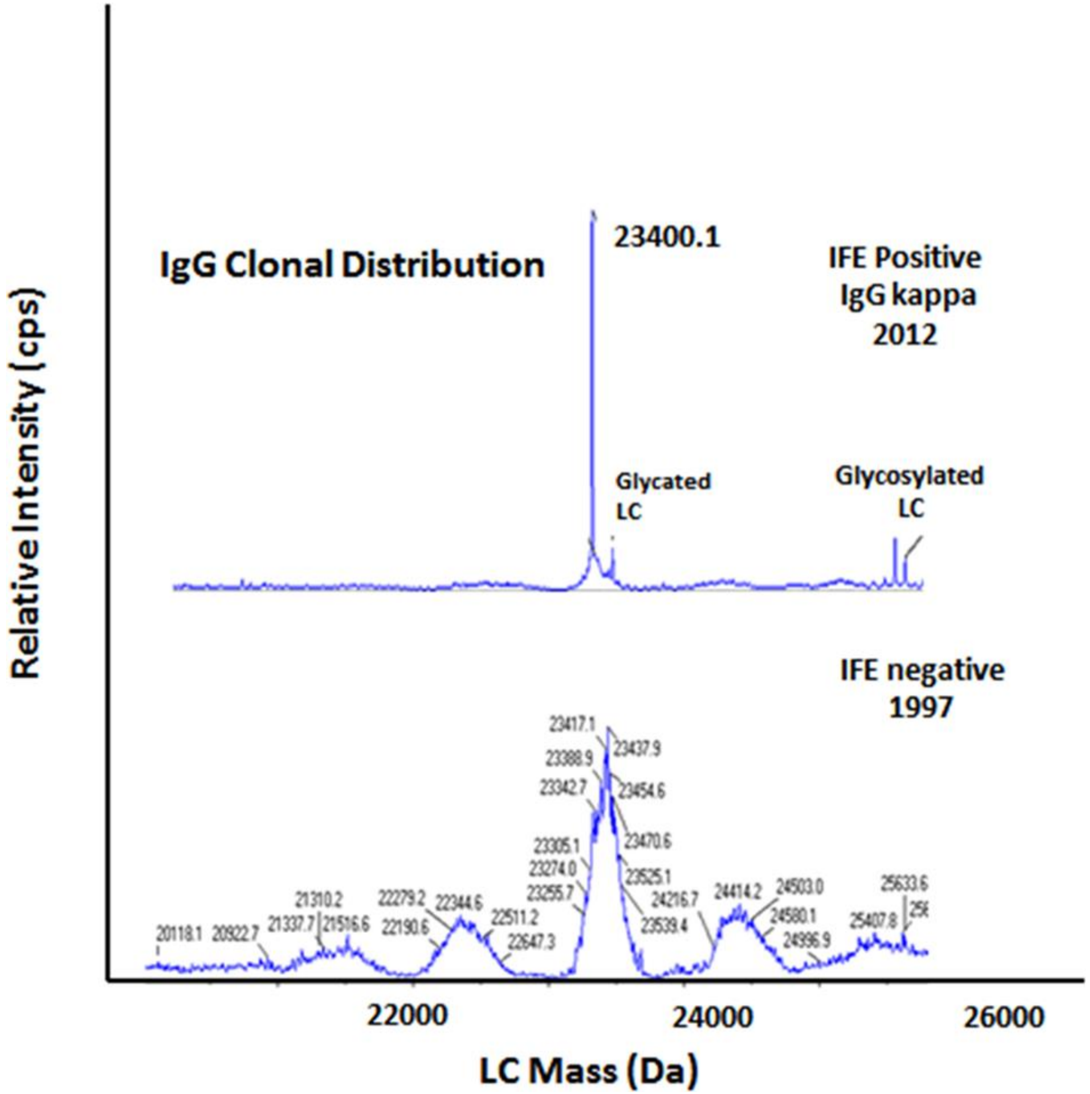
Serum Free Light Chains



Measuring the Abnormal Protein

Mass Spectrometry





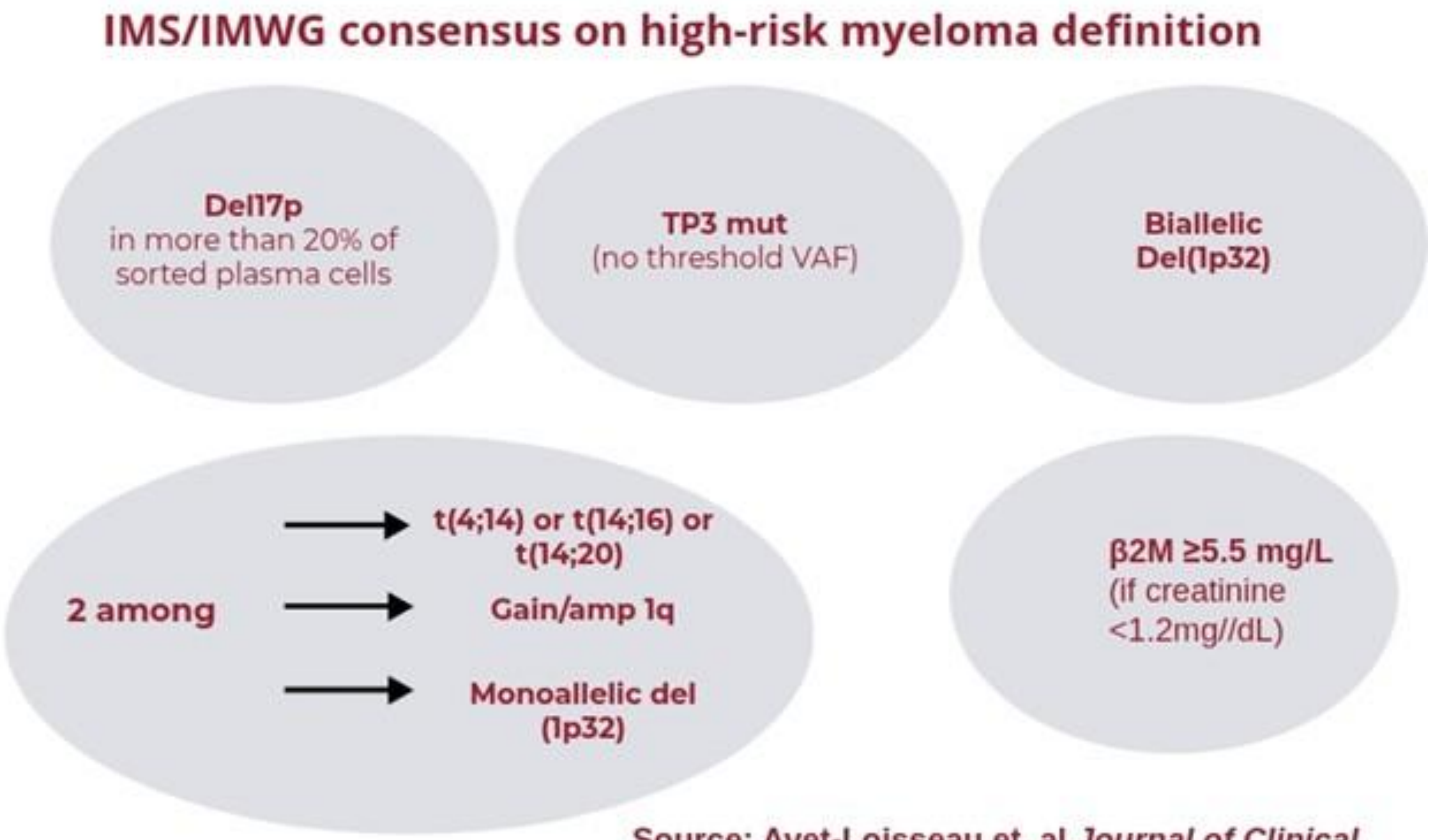
What Staging Means

A planning strategy, not a prognosis or life-expectancy

RISS	Blood Tests		LDH		Genetic "Mutations" (FISH)
I	Low Beta-2 Microglobulin Normal Albumin	+	Normal	+	Standard Risk
II	Intermediate values (Neither Stage I nor III)				
III	High Beta-2 Microglobulin	+	Elevated	O	High-Risk

New High-Risk Definition

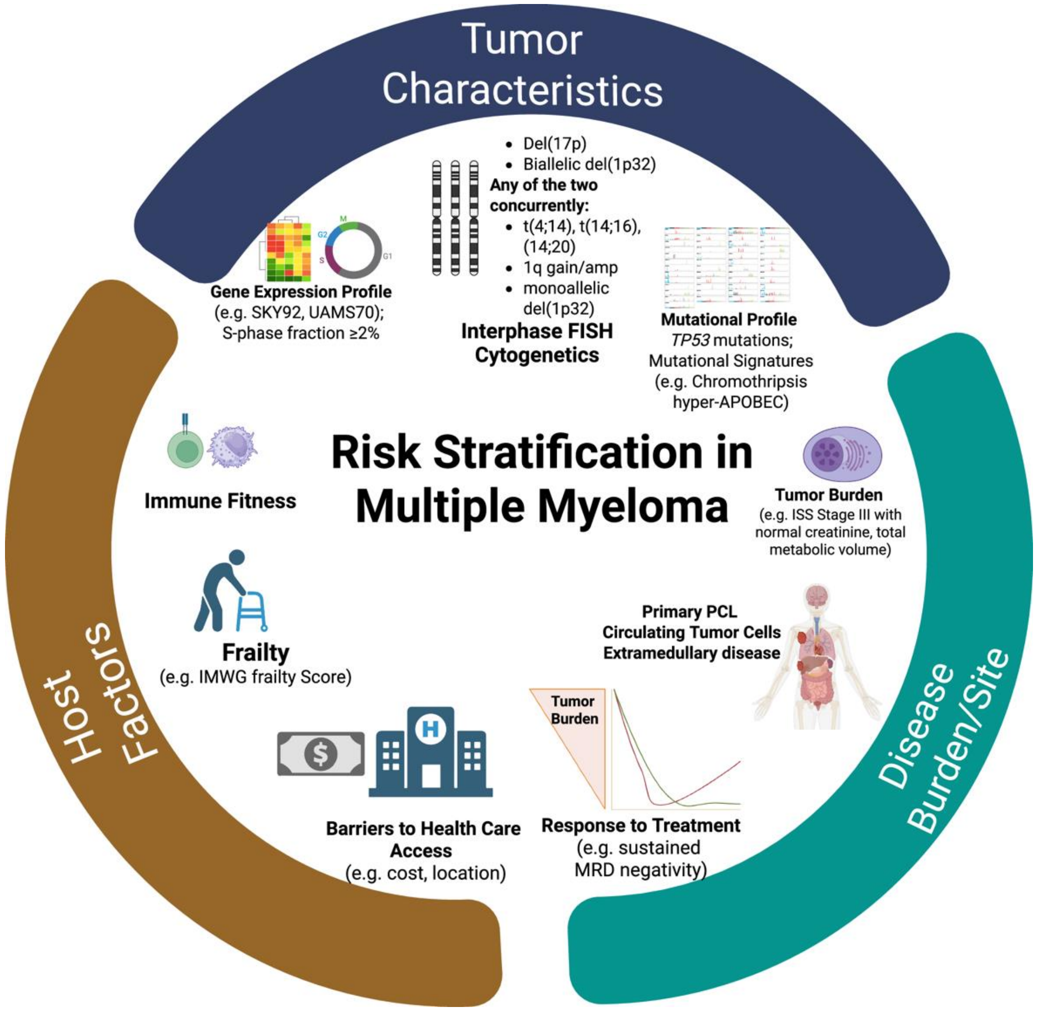
Again, useful to plan treatment



Source: Avet-Loisseau et. al *Journal of Clinical Oncology*

What It Means For Your Plan

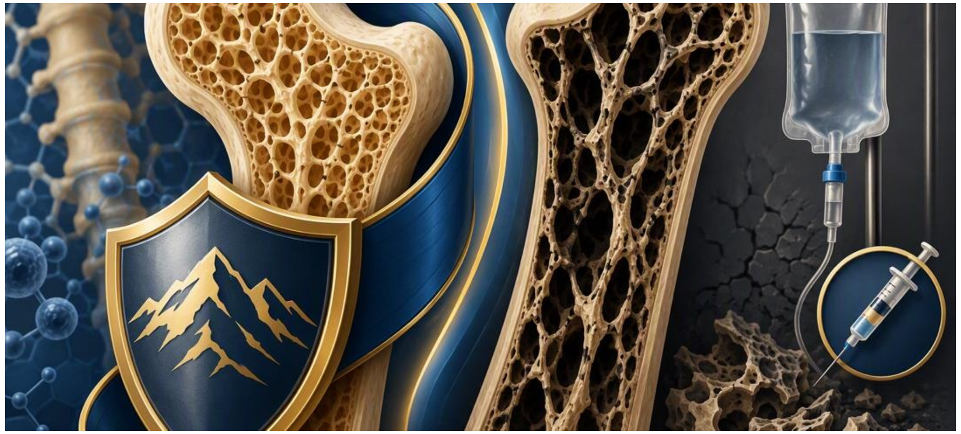
Different myelomas, different strategies



Bones

Protecting against fractures

- **Bone Strengthening Drugs:** Medications like Zoledronic Acid (Zometa) or Denosumab (Xgeva) protect bones from breakdown.
- **Fracture Reduction:** Regular bone-targeted therapy significantly lowers the risk of bone pain or fractures.
- **Dental Health Focus:** A dental check-up before starting bone strengthening drugs to avoid jaw-bone infections (ONJ).



New focal pain? Tell your team.

Infections

Roughly 7x the risk — prevention matters

Preventative Medications

Simple daily antiviral pills (like acyclovir) protect against shingles when taking targeted myeloma drugs like bortezomib or daratumumab. Depending on your treatment, you may need more, including an infusion of antibodies (IVIG).

Timely Vaccinations

Staying up to date with non-live vaccines (Flu, COVID-19, Pneumococcal) keeps your immune defenses ready throughout your treatment journey.

Early Fever Calls

Always report a fever right away. Early antibiotic treatment keeps minor infections from turning into major complications.

Neuropathy

Report numbness or tingling early

Signs to Notice Early

Peripheral neuropathy feels like tingling, numbness, or a "pins and needles" sensation in your toes or fingers. It is often linked to certain myeloma medications.

Quick Adjustments Help

Telling your team early allows us to tweak drug dose/schedule or even switch treatments, protecting your nerves before lasting changes occur.

Blood Clots

Know the warning signs = sudden chest pain, shortness of breath, leg swelling/pain

Why Risk Increases

Both myeloma itself and immunomodulatory medications (thalidomide, lenalidomide or pomalidomide) can temporarily increase blood clot risk.

Simple Daily Protection

Your doctor will prescribe a daily protective medicine—ranging from simple low-dose aspirin to blood thinners—tailored to your personal risk factors.

Whole-Person Wellbeing

Part of the plan, not an afterthought

Treatment focuses on your whole life, not lab numbers.

Caregivers are essential partners.

Exercise along with physical therapy if needed.

Open communication with your team.

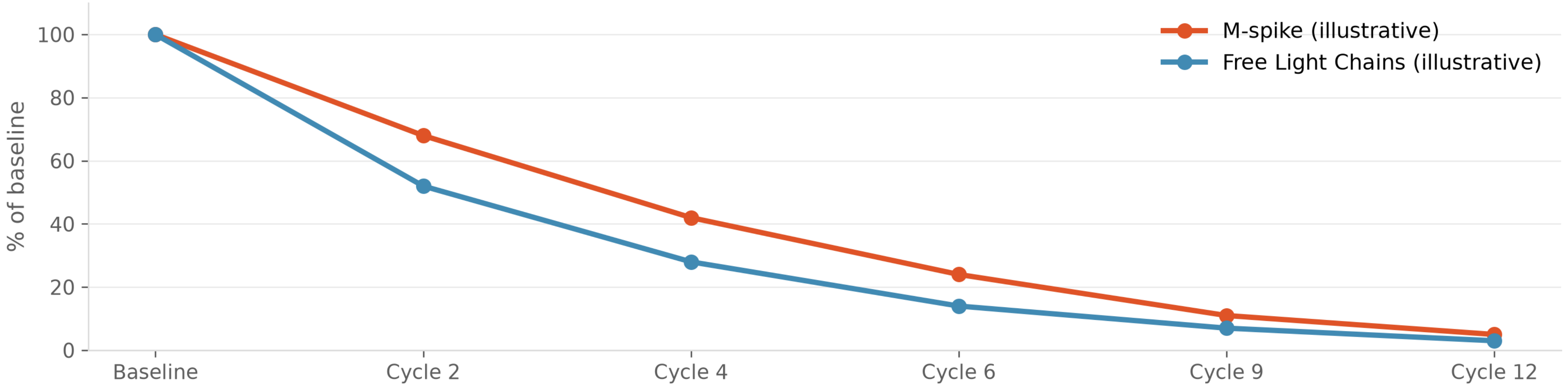
Good nutrition.

Emotional support.

What matters to you belongs in the plan.

Tracking Myeloma With Blood Tests

One value is a snapshot – the trend is the story.



Illustrative example only — not real patient data

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What “Remission” Really Means

Deeper Responses Usually Mean Longer Remissions

No myeloma cells in 100,000 - 1,000,000 cells

Minimal Residual Disease (MRD) Negativity

"Remission" means myeloma cells have dropped to very low levels. Modern tests like Next-Generation Flow and Next-Generation Sequencing can scan thousands to millions of bone marrow cells.

Why Free Light Chains Move Faster

An early hint of response

Intact Immunoglobulin (IgG, IgA, M-protein)

Half-Life: Weeks

Full antibody proteins clear from blood slowly. It takes several weeks to see significant drops on your SPEP lab report.

Free Light Chains (FLC)

Half-Life: Hours

Small light chain fragments clear very quickly.


Thank You!
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